Retail360

Predicting 60-Day Repeat Purchase

Can a customer's past purchases predict whether they will buy again?

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LIVE APP

Abstract

Retail analysts cannot review every customer. Our objective is to rank existing customers by whether they will buy again within 60 days, using only prior invoices. We compare basic RFM with richer purchase history and measure how many actual repeat buyers each model finds at the same review capacity.

Data Source

1,067,371 transaction rows

UCI Online Retail II · 2009–2011 · customer, invoice, product, date, quantity, price

Public app: 10 stored cases; no live scoring

Research Methods

01 COHORT + LABEL

Keep valid positive purchases. Eligible: ≥1 invoice in the prior 180 days. Label: ≥1 purchase in the next 60 days.

02 FEATURE FAMILIES

RFM → add recent orders, active days, order value and invoice gaps → add product variety. All features precede cutoff.

03 MODEL COMPARISON

Compare overall-rate dummy, logistic regression and histogram gradient boosting. Fit imputation and scaling on training snapshots only.

04 EVALUATION

Choose by validation AP. At later cutoffs, compare fixed models using average precision, buyers found at equal review capacity, and calibration.

Train: Sep & Dec 2010, Mar 2011 (n=9,659) → validate: Jun 2011 (n=2,659)

Exploratory checks only: Aug and Oct share 2,395 customers; no independent confirmation.

System Architecture

UCI transactions
Clean invoices

Python features
RFM · activity · variety

Prediction model
Validation-selected logistic

Streamlit app
Score · invoices · human review

Results

Prediction quality

September 2011 test · average precision (AP)

Overall rate	0.491
RFM	0.750
Full history	**0.755**

Temporal sensitivity · Full − RFM AP

Aug	+0.0074 [95% paired CI: 0.0022, 0.0128]
Sep	+0.0049 [95% paired CI: 0.0006, 0.0094]
Oct	+0.0046 [95% paired CI: 0.0002, 0.0088]

Review-capacity sensitivity

September 2011 · same number reviewed

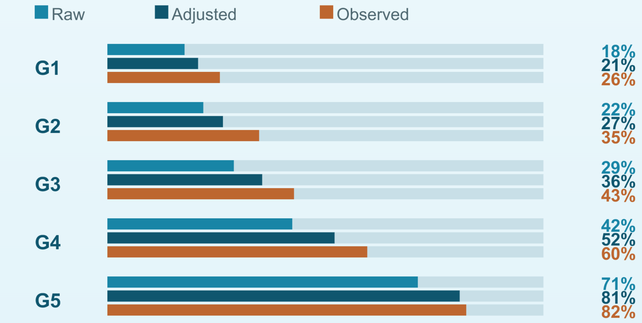
Top 5% · 139 review slots		
RFM 134		0
Full 134		
Top 10% · 278 review slots		
RFM 256		+2
Full 258		
Top 20% · 555 review slots		
RFM 458		-1
Full 457		

At 10%: only 2 more buyers among 278 reviewed.

At 20%: 1 fewer buyer. Aug/Oct at 10%: +1 each.

Probability reliability

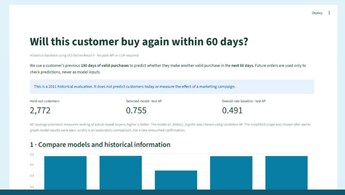
Sep 2011 · G1 lowest score, G5 highest



Brier: 0.222 → 0.209. All five groups still underpredict.

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Application & Evidence



Real customer 14397

Before cutoff: 9 invoices; last order 10 days earlier. Evidence: invoice 563982, source row 308453. Score 0.806 (top group; not exact odds). Returned in the next 60 days.

Conclusion & Next Step

Past purchases rank repeat buyers well. Richer features add little and the result changes with review capacity. Current scores support ranking and invoice review, not exact personal odds or campaign effects.

Next: test on a new, untouched period or retailer; improve calibration.

CORE FINDING Purchase history ranks buyers; extra features show little consistent practical gain.